RESERVOIR SIMULATOR

3D Reservoir Modeling & Production Optimization

DATABRICKS SOLUTION ACCELERATOR

Powered by Databricks Apps • Unity Catalog • Serverless

INTERACTIVE 3D RESERVOIR SIMULATION WITH AI-POWERED FIELD DEVELOPMENT PLANNING

Reservoir Simulator delivers an interactive 3D reservoir model with scenario management, well performance analysis, economic evaluation, and SAP ERP integration — calibrated to the Norne field (North Sea) benchmark dataset.

BUSINESS VALUE

- **Optimize $100M+ capital allocation** with scenario-based field development planning and NPV analysis
- **Visualize reservoir behavior in 3D** with oil/water saturation heatmaps, well trajectories, and layer navigation
- **Compare development scenarios** side-by-side with production, economics, and efficiency delta analysis
- **Integrate cost tracking** with SAP BDC for CAPEX/OPEX, procurement, and vendor contract management
- **Accelerate reservoir analysis** with Foundation Model API-powered AI assistant for optimization

CHALLENGES ADDRESSED

- **Expensive simulation software** — high-cost commercial tools with steep learning curves
- **Limited scenario comparison** — no easy way to compare multiple development plans side-by-side
- **Disconnected economics** — reservoir models and financial analysis in separate tools
- **No AI integration** — manual interpretation of simulation results without automated recommendations
- **Poor collaboration** — desktop-only tools preventing web-based team review of models

WHO IT'S FOR

Reservoir Engineer
3D visualization, scenario management, well results, and pressure tracking

Economics/Planning Analyst
NPV, IRR, payout period, cashflow analysis, and fiscal term modeling

Operations Manager
Field-level production summaries, well status, and operational KPIs

Development Geologist
AI-powered reservoir analysis and well placement optimization

5	1,000	9	40
Wells	Grid Cells	Dashboard Tabs	Timesteps

QUICK START GUIDE

1. **Clone Repository** — Clone the Reservoir Simulator repo to your local machine
2. **Build Frontend** — Run cd frontend && npm install && npm run build
3. **Deploy the App** — Run databricks apps deploy to launch the simulator
4. **Create Scenarios** — Configure wells, injection, and economic assumptions

GETTING THE MOST OUT OF RESERVOIR SIMULATOR

Explore 3D Reservoir
Navigate the Three.js viewport with saturation heatmaps, well paths, and layer views

Compare Scenarios
Run side-by-side comparison of multiple simulation runs with deltas

Analyze Economics
Configure price decks and fiscal terms for NPV, IRR, and cashflow analysis

Consult Reservoir Agent
Chat with the AI assistant for reservoir analysis and optimization

TARGET INDUSTRY

Upstream Oil & Gas — Reservoir engineering and field development planning for conventional and unconventional reservoirs. Calibrated to the Norne field (North Sea) benchmark.

⚠ **DEMO ONLY** — This application is a demonstration and solution accelerator. Please refer to the included LICENSE file for full terms of use.

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